CHRISTOPHER JOSHY

Software Engineer | AI Systems & Backend Development

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PROFESSIONAL SUMMARY

Computer Science Engineering student (B.Tech expected 2027) building AI-enabled, real-time, and full-stack systems. Developed student-facing products and healthcare platforms using Python, TypeScript, FastAPI, WebSockets, and cloud data services. Nexus Tech Lead experienced in coordinating student engineering teams, Git workflows, and code reviews. **Kaggle Silver Medal - NVIDIA Nemotron Model Reasoning Challenge; ranked 130 of 4,182 teams worldwide (top 3%).**

EDUCATION

B.Tech, Computer Science Engineering 2023-2027

St. Joseph's College of Engineering and Technology, Palai

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Lua; Rust (learning)

Backend & Real-Time: FastAPI, Node.js, Fastify, REST APIs, WebSockets, OAuth2/JWT

AI / ML: PyTorch, LLM orchestration, prompt engineering, OpenCV, MediaPipe, Wav2Vec2

Data & Infrastructure: MongoDB, Redis, Firebase, PostgreSQL, Docker, Git, Linux

PROJECTS

Typelo | Real-Time Competitive Typing Platform 2026

- Built synchronized head-to-head typing races with Redis-backed matchmaking, server-authoritative scoring, Elo ratings, bots, leaderboards, and anti-cheat checks for keystroke timing, sequence, and implausible speeds.

Stack: *React · TypeScript · FastAPI · WebSockets · Redis · MongoDB · Firebase*

Orix | College Ride Coordination Platform 2025-2026

- Built a verified-student ride-sharing platform with route/time/reputation-based matching, dynamic group formation, live group chat, safety reporting, and role-based administrative moderation.

Stack: *Flutter · FastAPI · Firebase Auth · MongoDB · Redis · WebSockets · Google Maps*

A.U.R.A. | Dementia Care Companion 2026

- Developed a patient-care ecosystem spanning a caregiver mobile app, FastAPI API, and Raspberry Pi edge module for face recognition, voice interaction, medication reminders, memory support, and SOS alerts.

Stack: *React Native · TypeScript · FastAPI · OpenCV · InsightFace · faster-whisper · MongoDB*

PROFESSIONAL EXPERIENCE

Nexus Tech Lead | SJCET Palai 2025-Present

- Lead institutional technical initiatives and coordinate student development teams; introduced Git workflows and code-review practices.
- Organize hackathons and workshops and mentor students in full-stack development and machine learning.

Software Engineering Intern | Pacelab, Kochi 2025

- Contributed within Agile/Scrum workflows through debugging, version control, and CI/CD-integrated testing.

ACHIEVEMENTS

- Kaggle Silver Medal - NVIDIA Nemotron Model Reasoning Challenge; ranked 130 of 4,182 teams worldwide (top 3%).
- 1st Place, IEEE CS SIGHT Society Website Competition; delivered a responsive, production-ready society website.

ADDITIONAL PROJECTS

Ghosted (ethical security testing) · Orito (multi-agent Linux assistant) · PROJECT H.Y.D.R.A. (satellite algal-bloom detection) · AI Maveli · ResumAI · GitDocGenAI · MaestraMind · WikiRunAI · Hando · Polaris Bot · KTU Notes Telegram Bot · ChrisTheTrader

AREAS OF INTEREST

Multi-agent orchestration · Machine learning infrastructure · Computer vision · High-performance backend systems · Distributed systems · Open source